

The importance of transparency: Declaring the use of generative artificial intelligence (AI) in academic writing

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Abstract

The integration of generative artificial intelligence (AI) into academic research writing has revolutionized the field, offering powerful tools like ChatGPT and Bard to aid researchers in content generation and idea enhancement. We explore the current state of transparency regarding generative AI use in nursing academic research journals, emphasizing the need for explicitly declaring the use of generative AI by authors in the manuscript. Out of 125 nursing studies journals, 37.6% required explicit statements about generative AI use in their authors' guidelines. No significant differences in impact factors or journal categories were found between journals with and without such requirement. A similar evaluation of medicine, general and internal journals showed a lower percentage (14.5%) including the information about generative AI usage. Declaring generative AI tool usage is crucial for maintaining the transparency and credibility in academic writing. Additionally, extending the requirement for AI usage declarations to journal reviewers can enhance the quality of peer review and combat predatory journals in the academic publishing landscape. Our study highlights the need for active participation from nursing researchers in discussions surrounding standardization of generative AI declaration in academic research writing.

INTRODUCTION

The integration of generative artificial intelligence (AI) into academic research writing has generated substantial hype within the

scientific community. AI-powered tools such as ChatGPT and Bard have garnered particular attention from academic researchers since the debut free research release of ChatGPT on 30 November 2022. These AI-powered tools are increasingly being utilized by

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researchers to assist them at various stages of the writing process of academic research. Generative AI tools assist authors in generating initial drafts by building upon authors' initial ideas and provide valuable suggestions for improving the existing text. With its contextual comprehension and human-like responses, generative AI tools present researchers with a fresh approach to idea generation, expanding the scope of research writing and fostering innovative thinking. The widespread adoption of generative AI tools has undoubtedly transformed the landscape of academic research writing.

While the potential benefits of generative AI tools in academic research writing are undeniable, ethical concerns have arisen alongside the hype. One key issue has sparked discussion within the academic research community is authorship and the attribution of AI-generated content. Research papers and scholarly works are attributed to human authors, reflecting their knowledge, expertise, and intellectual contributions. When employing generative AI tools, questions arise regarding how to appropriately disclose the AI system's role in the writing process. Recent debates among academic researchers suggest that crediting generative AI systems as the co-author of academic research articles is not suitable (Thorp, 2023), as these systems "do not fulfill the criteria for a study author, because they cannot take responsibility for the content and integrity of scientific papers." (Stokel-Walker, 2023) Several instances have emerged wherein ChatGPT has been included as a co-author in academic publications (O'Connor, 2023a, Flanagin et al., 2023). However, a subsequent wave of scrutiny has triggered a reconsideration of authorship attribution to generative AI tools (O'Connor, 2023b, Flanagin et al., 2023). Generative AI tools are nonlegal entities and are incapable of accepting responsibility and accountability for the content within the manuscript. Additional ethical concerns beyond authorship issues include copyright implications arising from the use of third-party content, conflict of interest, and the broader concept of plagiarism, which encompasses not only verbatim text copying but also the replication of ideas, methods, graphics, and other forms of intellectual output originating from others (Lund et al., 2023).

There is a growing consensus among academic research journals that transparency in declaring the use of AI in the research process is vital to uphold the integrity and credibility of academic research writing. Two notable initiatives in reporting research studies that involve AI intervention are CONSORT-AI (Consolidated Standards of Reporting Trials for Artificial Intelligence) (Liu et al., 2020) and SPIRIT-AI (Standard Protocol Items: Recommendations for Interventional Trials—Artificial Intelligence) (Cruz Rivera et al., 2020). The International Committee of Medical Journal Editors (ICMJE) recommended the need for authors to explicitly disclose the use of AI-assisted technologies, including large language models like ChatGPT, in the production of submitted work (ICMJE, 2023). Furthermore, efforts are underway to develop comprehensive reporting guidelines to evaluate the use of ChatGPT and large language models in scientific research and its impact on scientific research writing (Cacciamani et al., 2023). These guidelines aim to promote transparency in the research process by providing a framework for declaring the use of generative AI in academic research.

The importance of declaring the usage of generative AI tools by authors stems from two critical perspectives. One perspective relates to the phenomenon known as artificial hallucination, where generative AI models can produce text that appears coherent and meaningful but is actually fictional or lacks accuracy (Ji et al., 2023). Another well-known hallucination effect reported by some academic researchers is that generative AI tools sometimes include references of scientific studies that do not exist (Tam et al., 2023). This means that authors who use generative AI tools run the risk of including fabricated or inaccurate information in their research. These hallucinations can range from minor errors to entirely invented data that may seem plausible. By openly declaring the usage of generative AI tools, authors can transparently acknowledge the potential limitations and risks associated with the AI-generated content, thereby ensuring the integrity and credibility of their research. Another perspective is the potential bias in training data. Generative AI systems learn from large datasets, and if these datasets contain biases, the AI models can inadvertently reproduce and amplify them. Consequently, this can result in the generation of biased content that reinforces negative stereotypes and discrimination. When authors utilize generative AI tools in their research writing without disclosing it, there is a risk that these biases go unnoticed and unaddressed.

Our study aims to investigate whether nursing academic research journals require authors to declare their use of generative AI in their manuscripts. Examining whether nursing academic research journals require explicit declaration regarding the use of generative AI can offer insights into the level of transparency within the nursing research community. This holds implications for readers, allowing them to be well informed and make discerning judgments regarding the reliability, trustworthiness and credibility of content of the journals. The awareness of whether authors have utilized generative AI can aid readers in gauging the extent to which human expertise and machine-generated content contribute to the material.

This research effort not only sheds light on the extent to which authors in nursing journals acknowledge and disclose their employment of generative AI but also aims at initiating discussions on the responsible and transparent implementation of AI technologies in the realm of nursing research writing. By illuminating current practices and attitudes, our study contributes to the ongoing dialog about ethical considerations, disclosure protocols, and best practices when integrating AI-generated content into the academic publishing landscape of nursing studies.

MATERIALS AND METHODS

A list of nursing academic research journals was compiled by including all journals indexed in the Nursing category of the 2021 Journal Citation Reports—Science Edition (Clarivate, 2022). The Author Guidelines or Instructions to Authors for each of these identified nursing academic research journals were downloaded

on June 22–24, 2023. A thorough review of these guidelines was then conducted to ascertain whether they included any explicit statements or requirements concerning the utilization of generative AI in manuscript submissions. For Author Guidelines or Instructions to Authors that were written in non-English language, the Author Guidelines or Instructions to Authors were translated to English using Google Translate, and the translated version of the documents were reviewed. The aim of this review was to ascertain whether the guidelines contained any references to artificial intelligence. Using Google Translate for translation sufficed for achieving this specific objective. Descriptive statistics, including frequency and percentages, were used to summarize the results. Independent samples t-test was used to examine if there is any difference of the mean impact factor of the journal association between with and without explicit statements about the use of generative AI. Pearson's chi-square test of independence was used to examine the association between the themes of the journal (categorized according to the Scopus database) and the availability of the explicit statement.

RESULTS

A total of 125 nursing academic research journals were identified from the Nursing category of the 2021 Journal Citation Report and 121 of them published articles in English, two in Spanish, one in German and one in Italian. The Author Guidelines or Instructions to Authors of these journals were extracted. Among the four non-English journals, two had their Instructions to Authors available in English, whereas the other two did not. For the non-English journals lacking English versions of Instructions to Author, we employed Google Translate to translate their Instructions to Authors to English for subsequent evaluation. The Author Guidelines or Instructions to Authors were then reviewed by the authors. 47 out of 125 (37.6%) journals provided explicit statements about the use of generative AI tools in the writing process. 47 journals (37.6%) explicitly require declaration of the use of generative AI tools or AI-assisted technologies in the writing process, whereas 46 journals (36.8%) explicitly state that generative AI tools or AI-assisted technologies should not be listed as author or co-author. The result is included as [Supplementary Material S1](#).

The mean (SD) impact factors of those journals with and without explicit statement about the use of generative AI tools in the writing

process were 2.24 (0.82) and 2.15 (1.07), respectively; the mean difference (0.09, 95% CI from -0.27 to 0.45) was not statistically significant ($p=0.611$). There is no statistically significant association ($p=0.520$) between the themes of the journal (1. Specialized nursing journals, 2. General nursing journals, 3. Other nursing journals) and the availability of the explicit statement related to generative AI (Yes or No) ([Table 1](#)).

DISCUSSION

Exactly 47 out of 125 (37.6%) of nursing journals explicitly include information about utilizing generative AI tools in their Author Guidelines or Instructions to Authors on or before 24 June 2023, indicating that editorial boards of nursing journals are beginning to catch up on the ethical issues of using generative AI tools by authors in the writing process. We applied the same methodology to evaluate the Author Guidelines or Instructions to Authors from journals in the "Medicine, General and Internal" category. In about the same period, 25 out of the 172 journals (14.5%) in the "Medicine, General and Internal" category included information about the utilization of generative AI tools.

All of these 47 journals require the authors to declare generative AI tools usage in the body of the manuscript (e.g. in a separate declaration or acknowledgment section), rather than through paperwork or online form during the submission process to the editor. Mandating the inclusion of generative AI tool usage declaration within the manuscript facilitates clear communication to readers with essential information about the research process. This approach not only promotes transparency but also enables readers to evaluate the potential effect of generative AI tools on the article. We did not find any significant association between the Author Guidelines and (i) impact factors of the journals and (ii) categories of the journals.

There have been some initiatives in the declaration of generative AI usage in academic research writing among researchers (Cacciamani et al., 2023; Flanagan et al., 2023), but the authors are not aware of any published discussion of this issue in nursing studies journals. We are advocating for active participation from nursing researchers in these initiatives to ensure that our perspectives are included. We see a parallel between this declaration and the adoption of the Institutional Review Board (IRB) approval declaration in medical journals in the 1990s. This standardization emerged

TABLE 1 Crosstabulation of the journal category and statement of generative AI in the instruction of authors.

Category ^a	No	Yes	Total
Specialized nursing journals	39 (60.9%)	25 (39.1%)	64
General nursing journals	29 (60.4%)	19 (39.6%)	48
Other nursing journals	10 (76.9%)	3 (23.1%)	13
<i>p</i> -value from Pearson's chi-square test = 0.520	78 (62.4%)	47 (37.6%)	125

^aThe three categories were merged from smaller subgroups extracted from Scopus.



from a broader movement towards ethical oversight in human subjects research, spurred by key documents like the Belmont Report and the Declaration of Helsinki. Academic journals began to require IRB approval as part of the manuscript submission process, reflecting the growing emphasis on ethical oversight and the global trend towards standardized ethical practices in research (Amdur & Biddle, 1997). The declaration of using generative AI in academic research writing is a subject that is gaining attention and relevance in today's rapidly evolving technological landscape. Just as the declaration of IRB approval became a cornerstone of ethical research practice, the transparent acknowledgment of generative AI usage could similarly become a vital aspect of responsible academic conduct.

Another important aspect to consider is the declaration of the use of generative AI tools by journal reviewers. While authors are required to declare their utilization of generative AI tools in their submitted manuscripts, it is equally relevant to impose the same requirement on the reviewers. The declaration of generative AI tools usage by journal reviewers holds significant importance, particularly in the context of addressing the challenges posed by predatory journals. Predatory journals pose a threat to scientific integrity by charging authors "high article-processing fees but don't provide expected publishing services, such as peer review or other quality checks" (Singh Chawla, 2020). By openly declaring the use of generative AI tools, reviewers contribute to quality assessment, helping to differentiate reputable journals from predatory ones. This commitment to transparency reinforces the credibility of the review process and prevents inadvertent support for predatory journals. Additionally, the disclosure invites enhanced scrutiny and accountability, as predatory journals often lack robust peer review processes. Furthermore, explicit declaration helps expose deceptive claims of AI usage by predatory journals, safeguarding against manipulation and promoting transparency in publishing decisions for researchers.

CONCLUSION

Our study provides an understanding of the current landscape of declaring the use of generative AI tools by authors in nursing journals. Future studies could examine the implications of such disclosure on reader perception and the evaluation of research quality. This would help shed light on the effectiveness of disclosure in enhancing transparency and trust in academic writing. Furthermore, developing guidelines or best practices for the appropriate declaration of generative AI usage by authors and reviewers would be beneficial in ensuring consistent and responsible implementation of AI technologies in scholarly research.

CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Data S1.

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